Probing the deformed $N = 162$ subshell by the ^{238}U(^{40}Ar , $5n$)^{273}Ds reaction

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Abstract

A series of experiments was carried out at the SHANS-2 separator of IMP CAS to study the ^{238}U(^{40}Ar, $4n$–$5n$)^{274,273}Ds reaction. The excitation function was investigated in the excitation-energy range from 42 to 50 MeV. At 42 and 45 MeV, no correlated decay chain attributable to ^{273}Ds was identified, and upper limits of 0.34 and 1.09 pb were obtained. At 50 MeV, two complete decay chains of ^{273}Ds were observed, corresponding to a cross section of $0.14^{+0.18}_{-0.08}$ pb, in agreement with the value of $0.18^{+0.44}_{-0.12}$ pb reported previously for the same reaction at DGFRS-2. One chain follows the known high-energy, short-lived branch, whereas the other defines a lower-energy, longer-lived branch that supports the interpretation of a distinct state in ^{273}Ds. By combining the present data with earlier results, the population pattern and the α-particle energy distribution of ^{269}Hs suggest the presence of an additional low-energy state in this nucleus. The combined decay properties and isotopic systematics of Ds and Hs provide new constraints on the single-particle structure around the deformed $N = 162$ shell closure.

Keywords: α decay, $N = 162$ subshell, Isomeric state, Gas-filled recoil separator

1. Introduction

The stability of superheavy nuclei resulting from shell effects is one of the most intriguing subjects in nuclear physics. In particular, the deformed neutron shell gaps at $N = 152$ and $N = 162$ play important roles in the stabilization of the heaviest nuclei. At present, ^{252}Fm[1] and ^{270}Hs[2] are generally regarded as deformed doubly magic nuclei. Considerable progress has been made in understanding the structure of nuclei around $N = 152$ by decay properties [3–6], mass measurements [7–9], and spectroscopic studies of collective and single-particle excitations [10, 11]. In contrast, experimental knowledge of the low-lying level structure near $N = 162$ remains limited, primarily because of the extremely low production cross sections of nuclei in this region.

On the theoretical side, macroscopic–microscopic approaches have played an important role in establishing the shell structure of the superheavy nuclei. In this framework, a smooth macroscopic contribution to the nuclear energy is supplemented by microscopic shell and pairing corrections, with the single-particle spectrum commonly generated using a deformed Woods–Saxon potential. Calculations employing the Woods–Saxon potential together with higher-order deformation degrees of freedom predicted a pronounced region of enhanced stability around $Z = 108$ and $N = 162$ well before its experimental confirmation [12, 13]. Subsequent α-decay studies provided experimental evidence for the enhanced stability associated with the predicted $N = 162$ shell gap [14]. The successful description of this region makes the deformed Woods–Saxon framework particularly suitable for investigating the underlying single-particle structure around $N = 162$. Nevertheless, experimental data on the ordering and configurations of low-lying neutron single-particle states near the $N = 162$ subshell remain scarce.

To explore the level structure around the deformed $N = 162$ subshell, the ^{238}U(^{40}Ar,$5n$)^{273}Ds reaction was investigated in the present work. Investigation of low-lying single-particle orbitals in odd-A heavy nuclei provides a unique probe to the nuclear structure. Especially, $^{273}_{110}$Ds$_{163}$ and its α-decay daughter $^{269}_{108}$Hs$_{161}$, which represent one particle and one hole systems with respect to the $N = 162$ subshell closure, and thus offer direct access to the low-lying neutron single-particle structure around the $N = 162$ subshell. In this work, two complete decay chains were observed and assigned to ^{273}Ds on the basis of genetic correlations [15] and their consistency with previously reported decay properties.

The isotope ^{273}Ds was first identified at the GSI velocity filter SHIP as an α-decay daughter in cold-fusion reactions [16]. In the ^{208}Pb(^{70}Zn, n)^{277}Cn cold-fusion experiments at GSI [16, 17] and RIKEN [18, 19], the decay times attributed to ^{273}Ds were consistently of the order of 0.1 ms. More recently, direct production of ^{273}Ds through the ^{238}U(^{40}Ar, $5n$) reaction was reported by Oganessian *et al.* [20]. Two correlated decay chains were identified [15]. One chain agreed with previous data, whereas the other was assigned to an isomeric state of ^{273}Ds. Because both chains were incomplete, the decay properties of the daughter nucleus ^{269}Hs remained only weakly constrained. In the present work, the same reaction was studied at the SHANS-2 [21], and two additional complete decay chains assigned to ^{273}Ds were observed. These observations provide additional evidence for distinct decay patterns in ^{273}Ds and establish further correlations with the subsequent decays of ^{269}Hs, offering new constraints on the low-lying nuclear structure near the $N = 162$ subshell.

2. Experiment and Theoretical method

The experiment was performed at the China Accelerator Facility for Superheavy Elements (CAFE2)[22] of the Institute of Modern Physics by using the gas-filled recoil separator Spectrometer for Heavy Atoms and Nuclear Structure-2(SHANS-2)[21]. The nucleus ^{273}Ds was synthesized in the ^{238}U(^{40}Ar, 5n) fusion-evaporation reaction. A beam of $^{40}Ar^{13+}$ at energies of 217, 220, and 225 MeV in the laboratory frame was provided by CAFE2. The typical beam intensity on target was about 5 pμA, and the total effective irradiation time was about 728 h.

The targets were prepared by magnetron sputtering of ^{238}U onto Ti backings with thicknesses of 2.22 and 2.19 μg/cm^2, resulting in average ^{238}U layer thicknesses of 470 and 604 μg/cm^2. Ten arc-shaped target segments were mounted on a wheel with a diameter of 50 cm and rotated at 1500 rpm during irradiation in order to withstand the high beam load.

Evaporation residues were separated in flight from the primary beam and other reaction products by SHANS2. The separator was filled with helium gas at a pressure of about 1 mbar, and the magnetic rigidity was optimized for the transmission of ^{273}Ds. At the focal plane, the evaporation residues passed through two multi-wire proportional chambers (MWPCs) and were then implanted into a double-sided silicon strip detector (DSSD). The MWPCs were mounted about 20 cm upstream of the DSSD and operated with isobutane gas to distinguish the ER implantation from the decay events. The MWPC signals, together with the time-of-flight information, were used to distinguish implanted evaporation residues from subsequent decay events. The DSSD has an active area of 128×48 mm^2 and a thickness of 300 μm. The front side, defined as the X plane, is segmented into 128 vertical strips, whereas the back side, defined as the Y plane, is segmented into 48 horizontal strips. This geometry provides 6144 pixels. Six single-sided silicon detectors were arranged upstream of the DSSD in a box-like geometry to detect α particles and spontaneous-fission fragments escaping from the front side of the DSSD. Each detector has a thickness of 500 μm and an active area of 120×63 mm^2. In addition, three silicon detectors with a thickness of 300 μm were installed directly behind the DSSD as veto detectors in order to reject light particles punching through the focal-plane detector system. The entire silicon array was cooled to about -30°C with a circulating alcohol system to suppress thermal noise and improve the energy resolution. For full-energy α particles in the range from 5 to 9 MeV, the typical full width at half maximum of the DSSD was about 30 keV. The total detection efficiency of the silicon detector array for α particles was about 87(6)%. All detector signals were preamplified, shaped, and digitized with 14-bit analog-to-digital converters operated at 100 MHz in a self-triggered mode. Further details of SHANS2 and the detector system can be found in Refs. [21, 22].

The silicon detectors were calibrated with standard α sources (^{239}Pu, ^{241}Am, and ^{244}Cm), and with the well-known α-decay lines of ^{218}Rn, ^{214}Po, ^{216}Po, and ^{217}At. The reference α-particle energies were adopted from the National Nuclear Data Center (NNDC) NuDat database [23]. The separator transmission efficiency used for the cross-section evaluation was obtained from simulations performed according to the procedure described in Ref. [24], yielding a value of 20%. The experimental conditions and the measured cross sections are summarized in Table 1.

Table 1: Target thicknesses of ^{238}U, average projectile energies $\bar{E}_{\rm lab}$ at the middle of the target layers, corresponding average excitation energies $\bar{E}^*$ of the compound nucleus calculated from the mass tables of Refs. [25, 26], total beam doses, and measured cross sections σ or upper limits for the 4n and 5n evaporation channels in the ^{40}Ar + ^{238}U reaction. Values separated by "|" correspond to measurements with two target thicknesses, and the beam doses are listed in the same order.

Target thickness (mg/cm^2)	$\bar{E}^a_{\rm lab}$ (MeV)	$\bar{E}^*$ (MeV)	Beam dose ($\times 10^{19}$)	σ_{4n} (pb)	σ_{5n} (pb)
0.47 | 0.60	225	50	2.28 | 2.96	< 0.15	$0.14^{+0.18}_{-0.08}$
0.47	220	45	0.817	< 1.1	< 1.1
0.60	217	42	2.04	< 0.34	< 0.34

[a] The systematic uncertainty of the beam energy is 1 MeV.

To interpret the observed decay pattern, configuration-constrained potential-energy-surface (PES) calculations were performed in the deformation space ($\beta_2, \beta_4, \beta_6$). The nuclear surface is parameterized in terms of spherical harmonics, and the total energy is evaluated within the macroscopic–microscopic Strutinsky framework using a realistic phenomenological mean-field Hamiltonian [27–29]. In this approach, the total energy consists of a macroscopic contribution described by the standard liquid-drop model [30] and microscopic corrections arising from shell effects [31] and pairing correlations. The single-particle levels are generated from a deformed Woods–Saxon potential with the universal parameter set [32], while pairing correlations are treated using the approximate particle-number-conserving Lipkin–Nogami method [33], with the monopole pairing strengths G determined from the average-gap prescription [34]. For each specified one-quasineutron configuration, the corresponding unpaired neutron orbital is continuously tracked and blocked over the entire deformation lattice. The potential-energy surface is then obtained by minimizing the energy with respect to deformation while maintaining the assigned quasiparticle configuration, thereby allowing deformation and pairing effects to be treated self-consistently for each configuration.

3. Results and discussion

For the ^{238}U(^{40}Ar, xn)$^{278-x}$Ds reaction at an excitation energy of about 50 MeV, the 5n evaporation channel is expected to be favored [20]. At this energy, two decay chains were observed. The complete event information is presented in Fig. 1.

The energy and correlation time of the first α decay (α_1) in Chain 1 are consistent with those reported in previous cold-fusion ^{208}Pb(^{70}Zn, n)^{277}Cn experiments at GSI [16, 17] and RIKEN [18, 19]. The subsequent α_2 and α_3 decays, attributed to ^{269}Hs and ^{265}Sg, together with the terminal spontaneous-fission events, are also consistent with previously reported de-

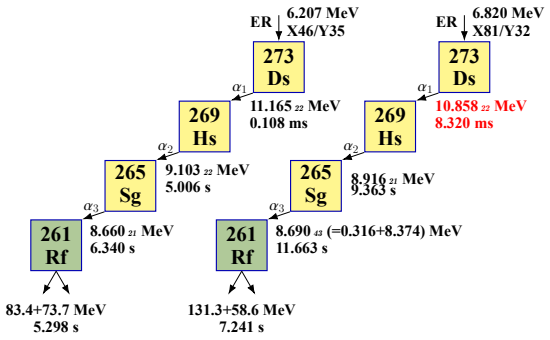


Figure 1: Two complete decay chains assigned to ^{273}Ds in the ^{238}U(^{40}Ar, $5n$)^{273}Ds reaction. For each chain, the energy of the evaporation residue and the corresponding X/Y implantation position are given at the top, followed by the energies of the successive α decays and of the terminal spontaneous-fission event together with the correlation times between adjacent signals. Energies in parentheses indicate summed signals. The numbers written as subscripts after the energies denote the corresponding uncertainties in units of keV. The red entries mark the newly observed branch and the related tentative state assignment discussed in the present work.

cay properties [2, 35–38]. The detailed assignments are discussed below.

The existence of two states in ^{261}Rf was suggested in earlier studies of the ^{277}Cn decay chains [16]. Düllmann and Türler subsequently reexamined the available decay data for ^{261}Rf and ^{265}Sg [35]. The decay properties of the two ^{261}Rf states, denoted ^{261}Rf^a and ^{261}Rf^b, were further investigated by Haba *et al.* [37, 38]. The longer-lived state, ^{261}Rf^a, has a half-life of 68(3) s and decays predominantly by α emission with E_α = 8.28 MeV. In contrast, ^{261}Rf^b has a half-life of $2.6^{+0.7}_{-0.5}$ s and decays predominantly by spontaneous fission, with a branching ratio of 82 ± 9% [38]. The terminal spontaneous-fission events observed in both chains, together with their correlation times, are consistent with the decay of ^{261}Rf^b.

The existence of two states in ^{265}Sg was proposed by Düllmann and Türler [35] and subsequently confirmed by Haba *et al.* [38] in the ^{248}Cm(^{22}Ne, $5n$)^{265}Sg reaction. The state ^{265}Sg^a has a characteristic α-particle energy of 8.84 ± 0.05 MeV and a half-life of $8.5^{+2.6}_{-1.6}$ s, with its α decay predominantly populating ^{261}Rf^a (91%). The state ^{265}Sg^b exhibits an α-particle energy of 8.69 ± 0.05 MeV and a half-life of $14.4^{+3.7}_{-2.5}$ s, and predominantly populates ^{261}Rf^b (80%). The measured α_3 energies and correlation times in both chains are consistent with the decay of ^{265}Sg^b. Together with the subsequent spontaneous-fission events, these observations favor the assignment of both chains to the decay sequence ^{265}Sg$^b \rightarrow {}^{261}$Rf^b.

Previous experiments on ^{269}Hs reported two α-particle energy groups at approximately 9.13 and 8.95 MeV [35, 36]. These energies are consistent with the observed α_2 energies in Chain 1 and Chain 2, respectively. The occurrence of two different α_2 energies followed by similar decay properties of ^{265}Sg suggests a nontrivial decay pattern involving low-lying states in ^{269}Hs. The possible nuclear-structure implications are discussed below.

The measured α-particle energies and correlation times are compared with previously reported data in Fig. 2.

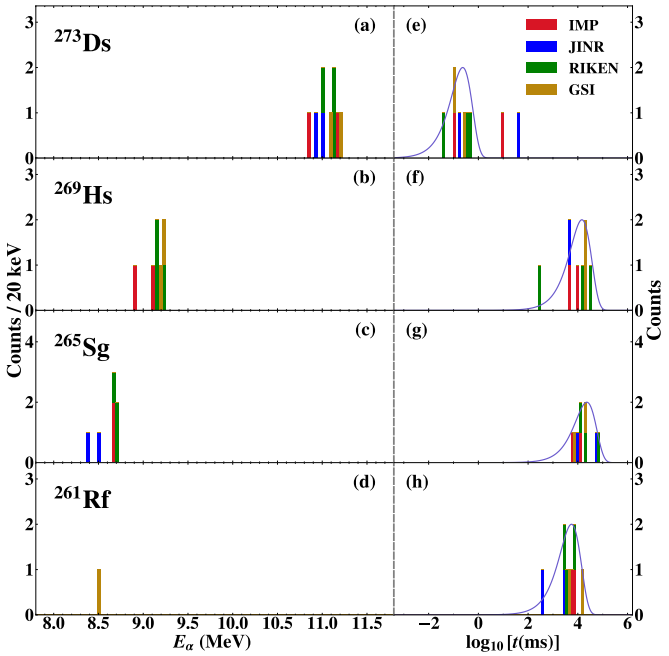


Figure 2: Compilation of the E_α values and decay-time distributions for the decay chains involving ^{273}Ds, ^{269}Hs, ^{265}Sg, and ^{261}Rf. Panels (a)–(d) show the measured α-particle energies, and panels (e)–(h) show the corresponding distributions of $\log_{10}[t(\mathrm{ms})]$. The red, blue, green, and yellow histograms represent the data from IMP, JINR [20], RIKEN [18, 19], and GSI [16, 17], respectively. The smooth curves in the decay-time panels are drawn according to Ref. [39].

3.1. *Decay properties of* ^{273}Ds

As shown in Fig. 2, most events in the two chains observed in the present work agree well with previously reported data. A notable exception is one ^{273}Ds decay event with E_α = 10.858(22) MeV and Δt = 8.320 ms, which differs substantially from the higher-energy, shorter-lived decays observed at GSI and RIKEN. Based on the reported half-life of $T_{1/2}$ = $0.18^{+0.11}_{-0.05}$ ms for the short-lived state [20], the probability of observing a decay time of $\Delta t > 5$ ms is only $P = 4.3 \times 10^{-9}$. This strongly suggests that the observed event originates from a different state in ^{273}Ds. Combined with the decay event reported in Ref. [20] [E_α = 10.929(22) MeV and Δt = 41.703 ms], the present observation supports the existence of a lower-energy, longer-lived branch in ^{273}Ds. Nevertheless, whether these two events originate from the exact same state remains uncertain.

The nucleus ^{273}Ds has been populated via two pathways: as the α-decay daughter of ^{277}Cn in the ^{208}Pb(^{70}Zn, n) cold-fusion reaction at GSI [16, 17] and RIKEN [18, 19], and by direct production in the ^{238}U(^{40}Ar, $5n$) hot-fusion reaction at JINR [20] and in the present work. Two distinct groups of decay properties have now been identified. Following Ref. [20], the lower-energy, longer-lived branch and the higher-energy, shorter-lived branch are denoted as ^{273}Ds^a and ^{273}Ds^b, respectively. The available data indicate that feeding from the α decay of ^{277}Cn selectively populates ^{273}Ds^b, whereas direct evaporation-residue production populates both branches. This population pattern is consistent with the assignment of ^{273}Ds^a to a distinct low-lying isomeric or ground state.

3.2. Decay properties of ^{269}Hs

Türler *et al.* [40] pointed out that the reported α-decay energies of ^{269}Hs in the range 9.17–9.25 MeV, observed as descendants of ^{277}Cn and ^{273}Ds, are among the highest known for this nuclide. Two possible explanations were mentioned: energy summing with conversion electrons or the presence of long-lived states in ^{269}Hs, similar to those established in ^{265}Sg and ^{261}Rf. However, the limited energy resolution and relatively large uncertainties of some previous measurements make it difficult to distinguish closely spaced α-decay branches. Therefore, rather than adopting specific state assignments from previous studies, we classify the compiled ^{269}Hs events into two empirical energy groups, denoted as ^{269}Hs^h and ^{269}Hs^l for the high- and low-energy groups, respectively, using E_α = 9.0 MeV as a practical dividing line. The event observed in the present work, with E_α = 8.915 MeV and τ = 9.363 s, is assigned to ^{269}Hs^l and represents the first lifetime determination for this energy branch.

As shown in Fig. 3, both ^{269}Hs^h and ^{269}Hs^l are populated in the directly produced ^{269}Hs sample. The corresponding event ratios are $n(^{269}\mathrm{Hs}^h) : n(^{269}\mathrm{Hs}^l) = 11 : 5$ for the directly produced sample and 6 : 1 for the sample populated through the α decay of ^{273}Ds. These ratios are based on the full event compilation shown in Fig. 3, including the literature event with an uncertain assignment. Notably, the ^{269}Hs^l event populated through ^{273}Ds decay is associated with its longer-lived branch, whereas the ^{269}Hs^h events are predominantly associated with the shorter-lived branch. No clear evidence for the cross-population channels ^{273}Ds$^b \rightarrow {}^{269}$Hs^l or ^{273}Ds$^a \rightarrow {}^{269}$Hs^h is found in the available data. Although the limited statistics prevent a definitive conclusion, this apparent selectivity may indicate the population of different low-lying configurations in ^{269}Hs through the two ^{273}Ds decay branches.

3.3. Discussion

Accordingly, the branch that starts from the higher E_α value and the shorter lifetime of ^{273}Ds is assigned to

$$^{273}\mathrm{Ds}^b \xrightarrow{\alpha} {}^{269}\mathrm{Hs}^h \xrightarrow{\alpha} {}^{265}\mathrm{Sg}^b \xrightarrow{\alpha} {}^{261}\mathrm{Rf}^b \xrightarrow{\mathrm{SF}} \text{fission}.$$

The branch that starts from the lower E_α value and the longer lifetime of ^{273}Ds is assigned to

$$^{273}\mathrm{Ds}^a \xrightarrow{\alpha} {}^{269}\mathrm{Hs}^l \xrightarrow{\alpha} {}^{265}\mathrm{Sg}^b \xrightarrow{\alpha} {}^{261}\mathrm{Rf}^b \xrightarrow{\mathrm{SF}} \text{fission}.$$

For clarity, the working assignments adopted in the present work are summarized in Table 2. On the basis of the present data, two distinct states are discussed for ^{273}Ds, denoted as ^{273}Ds^a and ^{273}Ds^b. Compared with ^{273}Ds^a and ^{273}Ds^b, the evidence distinguishing ^{269}Hs^h and ^{269}Hs^l as separate states rather than a single state remains inconclusive and relies solely on their differing decay properties. Because the statistics are limited and part of the historical data has a modest energy resolution. Therefore, this grouping is used simply as a basis for the tentative Nilsson configuration assignments discussed below, rather than as a firm experimental level scheme.

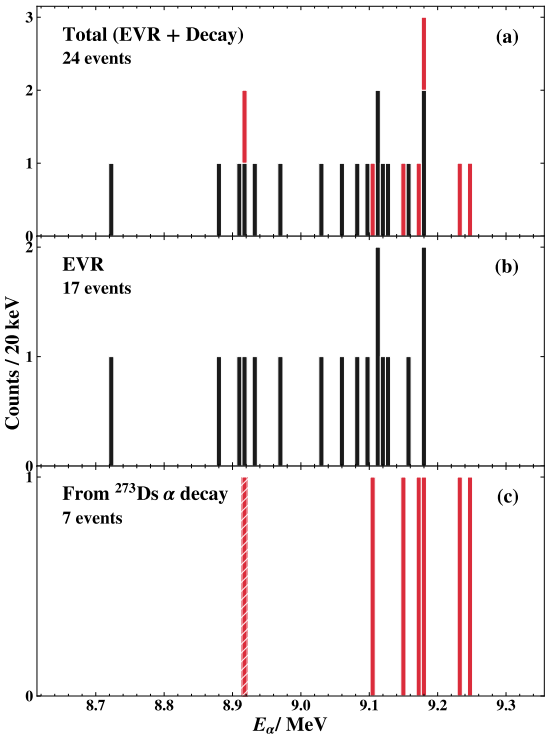


Figure 3: Measured α-particle energy distribution for events assigned to ^{269}Hs. Panel (a) shows the full event sample, panel (b) shows the directly produced evaporation-residue sample, and panel (c) shows the events populated through the α decay of ^{273}Ds. The hatched bar marks the low-energy event observed in the present work at E_α = 8.915 MeV.

Table 2: Tentative state labels used in the present work. The assignments define the working scheme adopted in the present discussion. The subdivision of the high-energy ^{269}Hs data into ^{269}Hs^a and ^{269}Hs^b follows the statistical analysis described in the text.

Nucleus	$T_{1/2}$	E_α^{a} (MeV)	Reference
^{273}Ds^a	$17.34^{+31.55}_{-6.83}$ ms[b]	10.86(2)	this work
^{273}Ds^b	$0.16^{+0.10}_{-0.04}$ ms	11.17(2)	this work,[16–20]
^{269}Hs^h	13^{+10}_{-4} s	9.20(4)	this work,[16–20]
^{269}Hs^l	$6.49^{+31.09}_{-2.97}$ s[b]	8.92(2)	this work
^{265}Sg^a	$8.5^{+2.6}_{-1.6}$ s	8.84(5)	Ref. [38]
^{265}Sg^b	$14.4^{+3.7}_{-2.5}$ s	8.69(5)	Ref. [38]
^{261}Rf^a	68 ± 3 s	8.28(2)	Ref. [38]
^{261}Rf^b	$2.6^{+0.7}_{-0.5}$ s	8.51(6)	Ref. [38]

[a] Energy uncertainties given in parentheses correspond to the data with the best energy resolution (68% confidence interval).
[b] Half-lives are calculated by Ref. [41].

In the neutron shell region between $N = 152$ and $N = 162$, the Nilsson orbitals $9/2^+[615]$, $7/2^+[613]$, $3/2^+[622]$, $1/2^+[620]$, and $11/2^-[725]$ are expected to be the most plausible configurations for low-lying states in ^{269}Hs [42, 43].

4. Summary and conclusions

A beam-energy scan of the ^{238}U(^{40}Ar, $4n$–$5n$)^{274,273}Ds reaction was performed at SHANS-2. No correlated decay chain attributable to ^{273}Ds was identified at excitation energies of 42 and 45 MeV, and upper limits of 0.34 and 1.09 pb were obtained. At 50 MeV, two complete decay chains of ^{273}Ds were observed, and the extracted cross section of $0.14^{+0.18}_{-0.08}$ pb agrees with the value reported previously for the same reaction at DGFRS-2.

One of the observed chains follows the known high-energy, short-lived branch of ^{273}Ds, whereas the other reveals a lower-energy, longer-lived branch. When the present data are combined with the literature results, the evidence supports the existence of two distinct decay branches in ^{273}Ds, and the lower-energy branch is naturally interpreted as a candidate low-lying or isomeric state.

The daughter nucleus ^{269}Hs shows a nonuniform population pattern. In addition to the previously recognized high-energy groups, the present work identifies a low-energy event at 8.915 MeV that is preferentially connected with the longer-lived branch of ^{273}Ds. Although the present statistics remain limited, this behavior is compatible with the presence of an additional low-lying state in ^{269}Hs.

The isotopic systematics of Q_α, $T_{1/2}$, and reduced α-decay width in the Ds and Hs chains provide further support for a structural change near $N = 162$. The overall picture is consistent with configuration-dependent α decay in odd-N transfermium nuclei and provides new constraints on the ordering of low-lying quasineutron orbitals around the deformed $N = 162$ shell gap. More decisive conclusions will require higher-statistics direct-production measurements and dedicated structure calculations.

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